

MODEL QUESTION PAPER

TED (15)6124

Reg.No;

(Revision 2015)

Signature

SIXTH SEMESTER DIPLOMA EXAMINATION IN WOOD AND PAPER TECHNOLOGY

PROCESS CONTROL AND INSTRUMENTATION

(Maximum marks: 100)

Time: 3Hrs

Marks

PART-A

(Maximum marks: 10)

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. State Sensitivity of an instrument.
2. Define Instrumentation
3. Define Absolute pressure, Atmospheric pressure,
4. Define Grammage .
5. List types of controllers in modern control system

(5X2=10)

PART.B

(Maximum Marks: 30)

II. Answer any five of the following questions. Each question carries 6 Marks

1. Describe the functional elements of measurement system with block diagram.
2. Explain on important instrumental errors.
3. Describe the working of Bourdon tube pressure gauge with diagrams.
4. Explain the measurement of gas density and specific gravity,
5. Explain PH scale and its application in process industry.
6. Explain pressure switch with neat diagram.
7. Explain Integral controller

(6x5=30)

PART.C

(Maximum Marks: 60)

(Answer one full question from each Unit.

Each Question carries 15 Marks)

UNIT.I

- III (a) Explain Pneumatic method of measuring force with a neat sketch 10
- (b) Describe Transducers. 5

OR

- IV (a) Describe
- (I) Indicators
- (II) Recorders. (2x4 =8)
- (b) Explain Electrical method of measuring force with a neat sketch 7

UNIT.II

- V (a) Describe Thermo couples with neat sketch 9
- (b) Describe Resistance temperature Detectors 6

OR

- VI (a) Explain Radiation type Pyrometer with neat sketch. List the advantages. 9
- (b) State method of measuring the speed. 6

UNIT.III

- VII(a). Explain the following types of moisture measurement
- (i) Infrared type and 4
 - (ii) Capacitance type 4
- (b) Explain freeenes, with neat sketch 7

OR

- VIII (a) Explain the following types of Measurement of consistency
- (i) Blade type 4
 - (ii) Rotary type 4
- (b) Explain Humidity measuremnt. 7

UNIT.IV

- IX. (a) Explain the following types of controls with its merits and demerits
- (i) Proportional,
 - (ii) Proportional derivative
 - (iii) Proportional integral (3X3=9)
- (b) Explain Manual and automatic control 6

OR

- X (a) Explain
- (i) Digital Computer.
 - (ii) Process Computer (2X5=10)
- (b) Lists the use of Analogue computer 5